

JUnit Testing Exercises

Exercise 1: Setting Up JUnit

Maven dependency (pom.xml):

```
<dependency>
  <groupId>junit</groupId>
  <artifactId>junit</artifactId>
  <version>4.13.2</version>
  <scope>test</scope>
</dependency>
```

Test class:

```
import org.junit.Test;
import static org.junit.Assert.*;

public class SetupCheckTest {

    @Test
    public void checkJUnitWorks() {
        assertTrue(true);
    }
}
```

OUTPUT:

```
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
```

Exercise 2: Writing Basic JUnit Tests

the class we want to test:

```

public class Calculator {

    public int add(int a, int b) {
        return a + b;
    }

    public int subtract(int a, int b) {
        return a - b;
    }

    public int multiply(int a, int b) {
        return a * b;
    }
}

```

test class:

```

import org.junit.Test;
import static org.junit.Assert.*;

public class CalculatorTest {

    Calculator calc = new Calculator();

    @Test
    public void testAdd() {
        assertEquals(8, calc.add(5, 3));
    }

    @Test
    public void testSubtract() {
        assertEquals(2, calc.subtract(5, 3));
    }

    @Test
    public void testMultiply() {
        assertEquals(15, calc.multiply(5, 3));
    }
}

```

Output:

```

Tests run: 3, Failures: 0, Errors: 0, Skipped: 0

```

Exercise 3: Assertions in JUnit

```
import org.junit.Test;
import static org.junit.Assert.*;

public class AssertionsTest {

    @Test
    public void testAssertions() {
        // Assert equals
        assertEquals(5, 2 + 3);

        // Assert true
        assertTrue(5 > 3);

        // Assert false
        assertFalse(5 < 3);

        // Assert null
        assertNull(null);

        // Assert not null
        assertNotNull(new Object());
    }
}
```

Output:

```
Tests run: 1, Failures: 0, Errors: 0, Skipped: 0
```

Exercise 4: AAA Pattern with Setup/Teardown

```

import org.junit.Before;
import org.junit.After;
import org.junit.Test;
import static org.junit.Assert.*;
import java.util.ArrayList;
import java.util.List;

public class ShoppingCartTest {

    private List<String> cart;

    @Before
    public void setUp() {
        // runs before every test method
        cart = new ArrayList<>();
        cart.add("Notebook");
    }

    @After
    public void tearDown() {
        // runs after every test method
        cart = null;
    }

    @Test
    public void testAddItem() {
        // Arrange
        String newItem = "Pen";

        // Act
        cart.add(newItem);

        // Assert
        assertEquals(2, cart.size());
        assertTrue(cart.contains("Pen"));
    }

    @Test
    public void testRemoveItem() {
        // Arrange
        String itemToRemove = "Notebook";

        // Act
        cart.remove(itemToRemove);

        // Assert
        assertEquals(0, cart.size());
        assertFalse(cart.contains("Notebook"));
    }
}

```

Output:

```
Tests run: 2, Failures: 0, Errors: 0, Skipped: 0
```